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Design and Development of a Smart Drunk Driving Detection and Intervention System Using Sensor Based Learning Anomaly Detection

A Thesis Presented to the Faculty of Computer Engineering
College of Engineering, Architecture and Technology
De La Salle University –Dasmariñas
Dasmarinas City, Cavite

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Computer Engineering

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CHAPTER 1

INTRODUCTION

This chapter discusses the background of the study, the objectives, the significance of the study, the scope and limitations, the conceptual framework, and the definition of terms of the study.

1.1 Background of the Study

Road safety remains a major concern worldwide, with drunk driving identified as one of the leading causes of traffic accidents. Alcohol is a substance that reduces the function of the brain, impairing thinking, reasoning and muscle coordination. All these abilities are essential to operating a vehicle safely. When drivers still proceed to drive even if they're under the influence of alcohol, their ability to respond to the road can lead to unexpected tragedies. It was said that approximately 40% of road accidents are caused by drivers being intoxicated by alcohol (Dairi et al., 2022). Because of this, people have tried to find a way to enforce safety regulations and find ways to prevent alcohol-related accidents from occurring.

There have been a lot of technologies developed to easily detect and discourage drunk driving. One commonly used device is the breathalyzer (DADSS, n.d); this device is used to measure the Blood Alcohol Content (BAC) via a breath sample. This tool is often used by police typically when a driver is suspected of driving intoxicated. Naturally, you must stop the driver from operating the vehicle to test their Blood Alcohol Content, which makes this a manual operation instead of it integrated into the vehicle. This causes the limitation of not always preventing the accident from happening in the first place. And since then, there has been a large number of demands in drunk detection applications in today's vehicles (Liu et al., 2022)

In the last few years, advancements in sensor technology and machine learning have opened new opportunities for developing intelligent drunk driving detection systems (Liu et al., 2022). Researchers have explored the use of multiple sensors,

such as gas sensors and temperature sensors, which can help in gathering data to help identify if the driver is intoxicated or not. And after that, Machine learning can help analyze the data given by the sensors to truly decide whether someone's drunk or not (Dairi et al., 2022). For example, anomaly detection approaches can be trained using normal driving data and later identify unusual patterns that may signal impaired driving behavior (Dairi et al., 2022). This way researchers don't only rely on the breathalyzer as a detection of potential intoxication.

Despite these technological developments, many existing studies primarily focus on detecting intoxication rather than implementing immediate preventive actions. Some systems can identify abnormal driver behavior but lack mechanisms that intervene, such as issuing alerts or communicating with external monitoring systems. Without these alerts, it may be insufficient preventing accidents from happening (Dairi et al., 2022).

To address this gap, there is a need to design a more comprehensive system where not only does it detect potential intoxication from the drivers but also provides immediate preventative action. Integrating multiple sensors with machine learning-based anomaly detection can provide reliable monitoring of driver conditions in real time. In addition, incorporating automated alerts and communication modules can help prevent potentially dangerous situations and notify relevant individuals or authorities when necessary. Therefore, this study aims to design and develop a smart drunk driving detection and intervention system using sensor data and machine learning techniques to enhance road safety.

1.2 Objective of the Study

General Objective

To design and develop a smart drunk driving detection and intervention system using sensor data and machine learning techniques to improve road safety and prevent alcohol-related driving incidents.

Specific Objectives

1. To collect and analyze driver-related sensor data, including gas sensor readings, temperature data, and camera inputs.
2. To develop and apply an anomaly detection model capable of identifying abnormal patterns that may indicate drunk driving.
3. To implement a real-time alert mechanism that provides visual and audio warnings when potential intoxication is detected.
4. To design a communication module that sends notifications or status reports to designated contacts or monitoring systems when drunk driving is detected.

1.3 Significance of the Study

Drivers. This study will help drivers become more aware of the risks associated with drunk driving. The proposed system can detect possible intoxication using sensor data and provide immediate visual and audio warnings. This will encourage drivers to make safer decisions before operating a vehicle and reduce the likelihood of accidents caused by alcohol

Passengers and Road Users. Passengers, pedestrians, cyclists, and other motorists may benefit from the improved road safety provided by the system. By detecting intoxicated driving and activating preventive measures such as ignition restriction or speed limitation, the system can reduce the chances of road accidents and help protect people who share the road.

Traffic Law Enforcement. The findings of this study may provide useful information for authorities and organizations working to improve road safety. The proposed system can support existing traffic safety programs by offering a technological solution that helps detect and prevent drunk driving incidents.

Vehicle Manufacturers. This study may contribute to the development of smarter vehicle safety technologies. The integration of sensors, machine learning, and automated intervention mechanisms can serve as a reference for future vehicle designs that aim to enhance driver monitoring and accident prevention systems.

Future Researchers. This study may serve as a reference for future researchers who wish to explore intelligent transportation systems, machine learning applications in driver monitoring, or sensor-based safety technologies. The findings may help them improve, expand, or develop new approaches to detecting and preventing drunk driving in real-world environments.

1.4 Scope and Limitations

This study focuses on the design and development of a Smart Drunk Driving Detection and Intervention System that uses sensor data and machine learning techniques to identify possible intoxicated driving. The system utilizes gas sensors, temperature sensors, and camera inputs to collect data related to the driver and the surrounding environment. These data are processed using dimensionality reduction (t-SNE) and an Isolation Forest anomaly detection algorithm to identify abnormal patterns that may indicate drunk driving. When intoxication is detected, the system generates visual and audio alerts, activates vehicle control mechanisms such as ignition restriction or speed limitation, and sends notifications to designated contacts or monitoring systems.

The study is limited to the development and testing of the detection and intervention system in a controlled prototype environment. It focuses on evaluating the capability of the system to detect abnormal patterns based on sensor data and to trigger automated responses. The research does not cover full-scale deployment in commercial vehicles or integration with all types of vehicle systems. Additionally, the system primarily relies on sensor readings and algorithm-based analysis, which may be affected by environmental conditions, sensor accuracy, or variations in driver behavior.

These delimitations are set to ensure that the study remains focused on the design, implementation, and evaluation of the proposed drunk driving detection and intervention system, while acknowledging factors that are beyond the control of the researchers.

1.5 Conceptual Framework

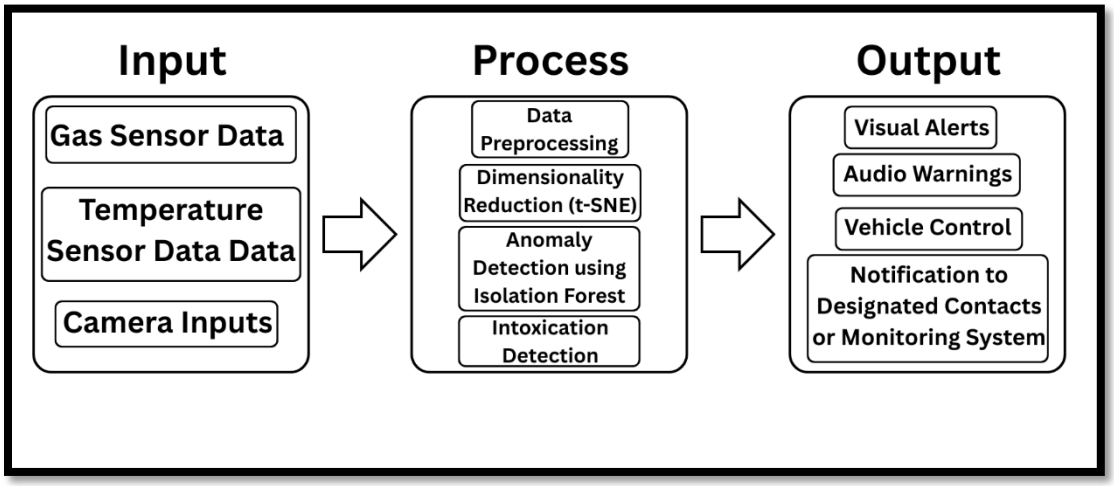

Figure 1.1 Conceptual Framework

1.6 Definition of Terms

Anomaly Detection - anomaly detection refers to a method used to identify unusual patterns or behaviors that do not conform to expected data patterns. In this study, anomaly detection refers to the use of the Isolation Forest algorithm to detect abnormal sensor data that may indicate drunk driving.

Isolation Forest Algorithm - a machine learning algorithm used to identify anomalies within a dataset. In this study, it refers to the algorithm applied to sensor data to determine whether the detected patterns deviate from normal driving conditions.

Machine Learning - a field of computer science that enables systems to learn patterns from data and make predictions or decisions. In this study, machine learning refers to the technique used to analyze sensor data and detect potential intoxicated driving behavior.

t-SNE (t-Distributed Stochastic Neighbor Embedding) - a machine learning technique used for dimensionality reduction and visualization of high-dimensional data. In this study, t-SNE refers to the method used to extract meaningful patterns from collected sensor data before applying anomaly detection.

References

A. Dairi, F. Harrou, & Y. Sun. (2022). Efficient driver drunk detection by sensors: A manifold learning-based anomaly detector. IEEE Access, 10, 121345–121356. https://doi.org/10.1109/ACCESS.2022.3219425

DADSS (n.d). *Can technology stop drunk driving?* Dadss - Driver Alcohol Detection System. https://dadss.org/news/updates/can-technology-stop-drunk-driving/#:~:text=Sensing%20impairment,from%20where%20we%20are%20today.

Liu, J., Luo, Y., Ge, L., Zeng, W., Rao, Z., & Xiao, X. (2022). An Intelligent Online Drunk Driving Detection System Based on Multi-Sensor Fusion Technology. *Sensors (14248220)*, *22*(21), 8460. https://doi.org/10.3390/s22218460